



AI Lab VPS

Learning · Building · Publishing (GitHub + HF)

This document is the master plan for a personal AI engineering laboratory hosted on an OVHcloud VPS. The goal is to learn Linux administration, Docker, backend development and agentic AI by building real projects — publishing each one publicly on GitHub and Hugging Face as a portfolio. The infrastructure is built in phases (0–4), while a 5-month roadmap guides the studies; every month ends with a published demo, a journal entry and an updated roadmap.



Infrastructure — Phase 0–4

Phase 0 — Prep

- ☐ OVH edge firewall: allow 22/80/443 only
- ☐ SSH from Mac (id_ed25519)
- ☐ Rescue mode instructions noted

Rescue: OVH dashboard → VPS → Reboot into rescue mode (netboot) → connect via SSH to rescue IP w/ password → mount /dev/sda1 (or vg) → recover. Never lose root/debian password while it's stored in Infisical.

Phase 1 — Server Foundation

- ☐ apt update + upgrade (Debian 13)
- ☐ TZ America/Sao_Paulo · hostname ai-lab · timesync
- ☐ qemu-guest-agent installed (consistent OVH snapshots)
- ☐ swapfile 2 GB + swappiness=10
- ☐ unattended-upgrades (security only)
- ☐ user bruno (sudo) created
- ☐ SSH key copied + login verified (2nd session)
- ☐ sshd hardened: root+password OFF (sshd -t → reload)
- ☐ passwd -l root (OVH email password dead)
- ☐ Fail2Ban SSH jail
- ☐ UFW 22/80/443 + DOCKER-USER default-drop (Phase 2)

Phase 1.5 — Systemd-first (learn Linux before Docker)

- ☐ PostgreSQL via apt installed
- ☐ FastAPI "hello" as systemd unit + gunicorn
- ☐ journalctl / systemctl mastered
- ☐ app containerized in Phase 3 (same app)

Phase 2 — Docker

- ☐ Docker Engine (official repo) + Compose plugin
- ☐ bruno added to docker group

- ☐ daemon.json log rotation (10m × 3)
- ☐ DOCKER-USER default-drop persisted (survives reboot)
- ☐ hello-world test passed

Phase 3 — Slim Stack (only what's studied now)

- ☐ PostgreSQL 16 container (healthcheck · mem_limit)
- ☐ Caddy reverse proxy (only 80/443 published)
- ☐ pinned image tags · restart: unless-stopped
- ☐ internal-only network (nothing else exposed)
- ☐ Systemd FastAPI moved into container
- ☐ Secrets via Infisical: `infisical run -- docker compose up -d` (no .env on server)

Phase 4 — Backup & Resume

- ☐ scripts/backup.sh: pg_dump + volume tar (age/gpg)
- ☐ cron nightly (user bruno)
- ☐ rclone → Backblaze B2 / OVH Object Storage
- ☐ Monthly restore drill (OVH mount option)
- ☐ Pause/resume runbook in README (recreate in minutes)



Learning Roadmap — every month ends with a PUBLISH step

Month 1 — Linux + Docker

- **Study:** HF Transformers · HF Datasets · HF Spaces
- **Build:** first chatbot Space
- **Publish:** Space (live demo) + GitHub repo + journal entry + ROADMAP update
- ☐ Transformers: pipeline(), AutoTokenizer, AutoModelForCausalLM
- ☐ Datasets: download, filter, publish one
- ☐ Chatbot Space duplicated + modified + public
- ☐ Repo flipped public (when polished)

Month 2 — Backend

- **Study:** HF Inference API (huggingface_hub) · SmolAgents · Redis 7
- **Build:** agent demo (calls APIs, executes code)
- **Publish:** Space + repo
- ☐ Inference API: swap models without changing code
- ☐ SmolAgents: agent with tools + ReAct loop
- ☐ Redis 7 added (cache/queues when app needs it)
- ☐ FastAPI REST + auth + async mastered

Month 3 — AI Integration

- **Study:** OpenRouter · Evaluation (BLEU · ROUGE · BERTScore · LLM-as-Judge)
- **Build:** LLM Evaluation Lab (FastAPI on VPS)
- **Publish:** eval UI Space + repo + dataset (results/prompts)

- ☐ Models via OpenRouter + HF (multiple providers)
- ☐ Metrics implemented + stored in PostgreSQL
- ☐ Eval dataset published on HF
- ☐ Dashboard/UI for comparing models, cost, latency

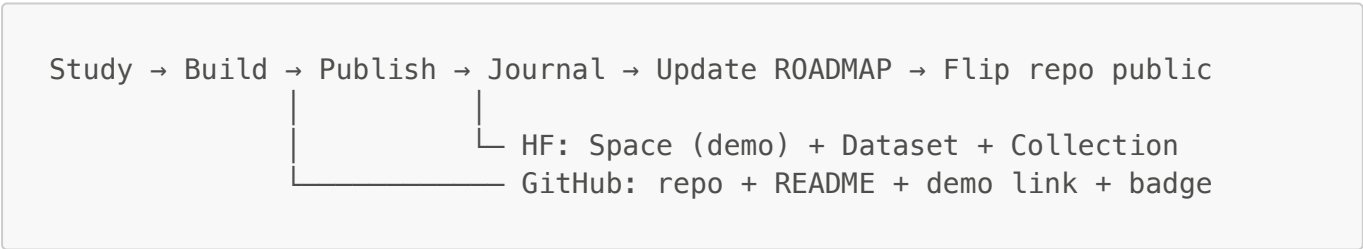
Month 4 — Agentic AI

- **Study:** MCP · RAG · Embeddings · Qdrant
- **Build:** MCP Playground + Agent Evaluation Lab
- **Publish:** agent lab Space + repo
 - ☐ MCP server: tools + resources (agent ↔ tools)
 - ☐ Qdrant added: embeddings + semantic search
 - ☐ RAG pipeline end-to-end (docs → vectors → answers)
 - ☐ Agent eval: planning, tool use, memory, reliability

Month 5 — Professional

- **Study:** Langfuse Cloud · CI/CD · Monitoring · Portfolio
- **Build:** portfolio (projects + demos unified)
- **Publish:** everything
 - ☐ Langfuse Cloud wired (prompts, costs, latency, errors)
 - ☐ CI/CD pipeline (tests + deploy on push)
 - ☐ All 5 project repos public + READMEs + demos
 - ☐ ROADMAP 100% complete · HF profile pinned · GitHub grid full

 Publish Loop (the habit) — repeat every month



 Backup — 3 Layers

Layer	Where	What
Code	GitHub	compose · Caddyfile · scripts · migrations
Config	Infisical (cloud)	secrets — never on server, never committed
Data	Backblaze B2 / OVH Object Storage	encrypted (dumps + volumes)

Secrets (Infisical): API keys and passwords live in Infisical Cloud, injected at runtime with `infisical run`. No `.env` files on the server or in Git — removes the top secret-leak risk and simplifies restore (no secrets to back up).

 Machine Config — OVHcloud VPS-1

Provider	OVHcloud VPS-1 2027
Plan	VPS Local Storage
CPU	2 vCPU
RAM	4 GB
Disk	40 GB SSD (local storage)
OS	Debian 13 (Trixie)
Backup	OVH Automated Backup (1 rotation) — daily, included
Billing	Monthly — 4.49 €/month (with coupon)
Region	Canada East (Beauharnois) — good latency from Brazil
IPv4	Included